

Max-Flow Problem

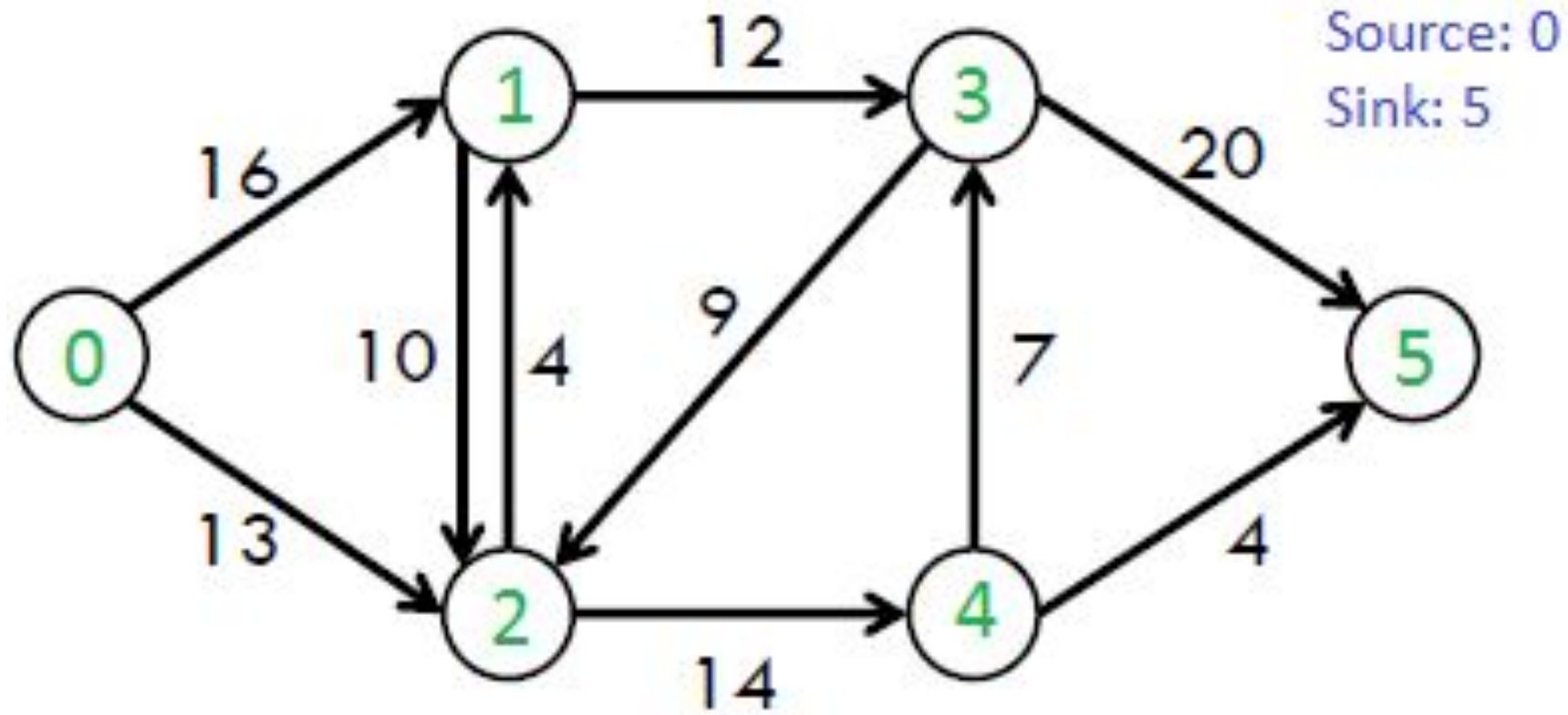
Max-Flow Problem

Given a directed graph where each edge has a capacity each edge has a capacity $c(u,v) \geq 0$, a source node s , and a sink node t , find the greatest possible amount of flow that can be sent from s to t , satisfying the following constraints:

- Capacity constraint: $0 \leq f(u,v) \leq c(u,v)$
- Flow conservation: For every vertex v except s and t :

$$\sum_u f(u,v) = \sum_w f(v,w)$$

Max-Flow Problem



Ford-Fulkerson Algorithm

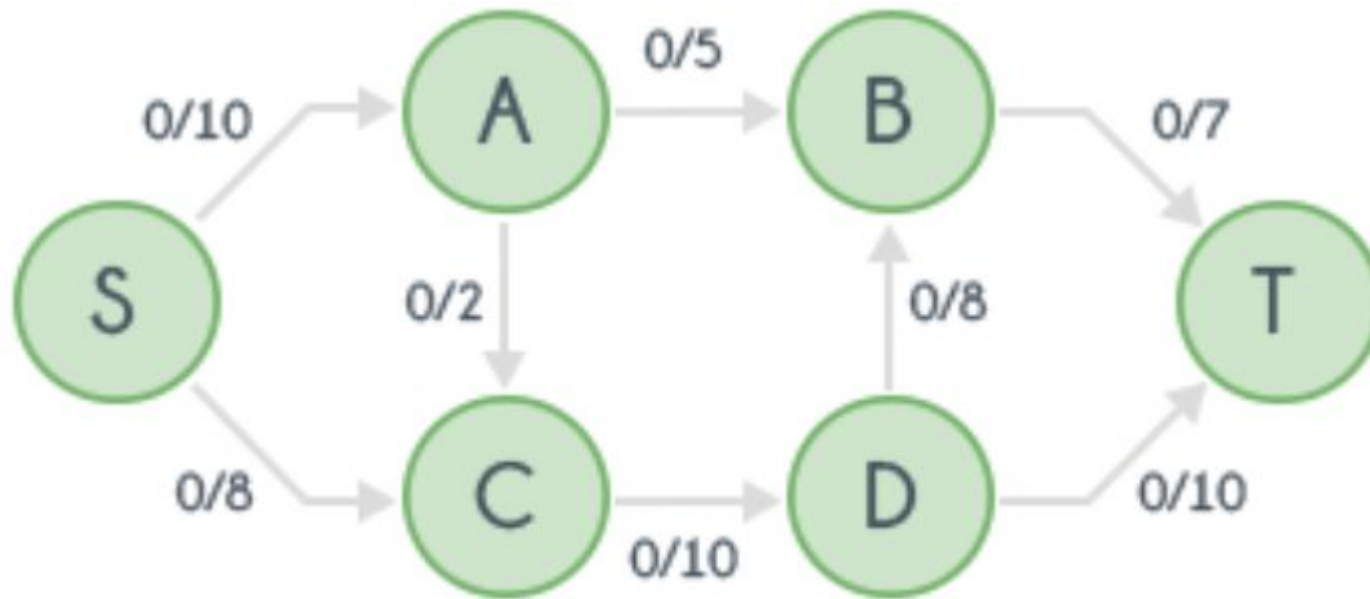
Need to know:

- Residual Graph
- Augmenting path

Ford-Fulkerson Algorithm

- Initialize maxflow as 0
- While there is augmenting path
 - Determine the flow possible through the augmenting path
 - Add that flow to the maxflow
 - Find the residualGraph

Practice



Edmond-Karp Algorithm

- Ford-Fulkerson is a more general method that doesn't specify how to find augmenting paths.
- Edmonds-Karp is a specific implementation that uses Breadth-First Search (BFS) to find the shortest augmenting paths (in terms of number of edges, not the sum of weights of edges).

Max Flow Problem

- <https://www.geeksforgeeks.org/ford-fulkerson-algorithm-for-maximum-flow-problem/>
- <https://brilliant.org/wiki/ford-fulkerson-algorithm/#:~:text=The%20Ford%2DFulkerson%20algorithm%20is,data%20through%20a%20computer%20network.>
- https://cp-algorithms.com/graph/edmonds_karp.html#flow-network